Questions

Q1.

(i) Describe **one** situation where separation of electric charge can create a spark.

(2)

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(ii) In a spark, the total charge of 0.22 μC (microcoulombs) flows in 2 ms (milliseconds).

Calculate the average current in that time.

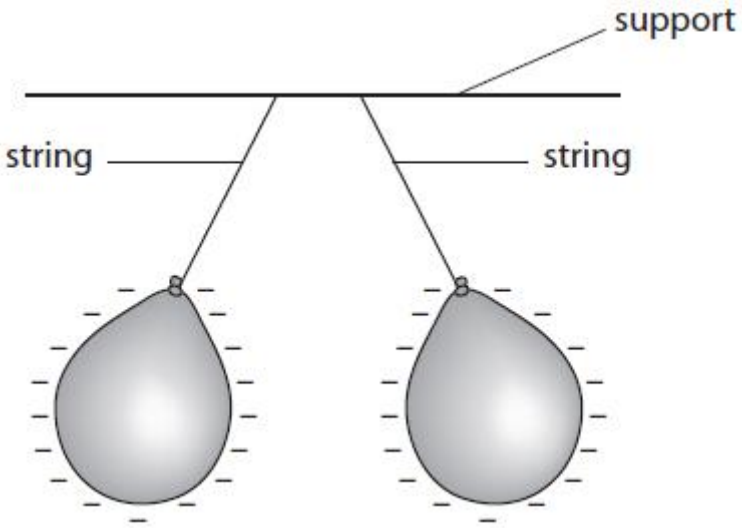
(4)

average current = ... A

(Total for question = 6 marks)

Q2.

(a) A student ties two balloons to a support with some string.
The student rubs both balloons with a dry cloth which gives the balloons a negative charge.
The diagram shows the balloons after they were rubbed.



Use words from the box to complete the sentences.

(4)

attract	charge	electrons	negative	neutral
neutrons	positive	protons	repel	support

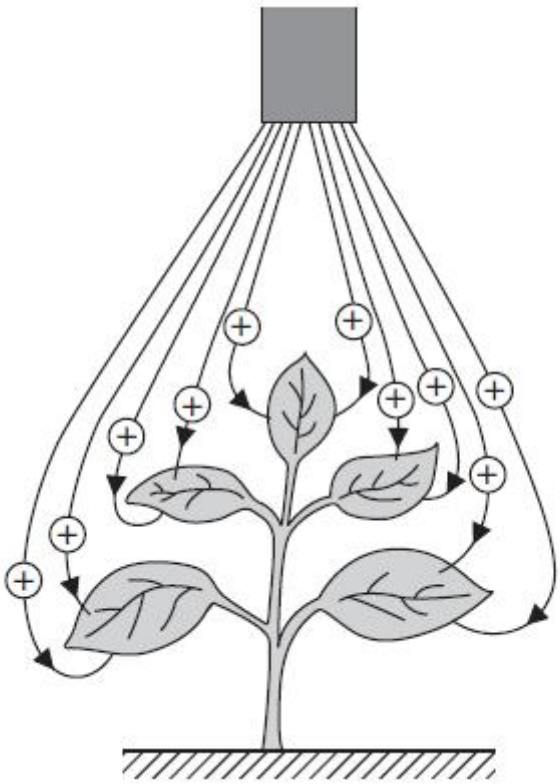
The balloons ..... each other because they have the same .....

The cloth is left with a ..... charge.

The charged particles that are transferred from the cloth to the balloons are called .....

(b) The diagram shows an electrostatic insecticide spray being used on a plant. The plant is initially uncharged.

Each droplet of spray is given a positive charge.



(i) Explain the advantages of using an electrostatic insecticide spray compared to an uncharged

insecticide spray.

(3)

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(ii) There is a current of 0.008 A in the sprayer for a time of 10 minutes.
Calculate the charge supplied to the sprayer in this time.

(3)

charge = ... C

(Total for Question = 10 marks)

Q3.

A plastic rod and a piece of cloth are both uncharged.

A student rubs the plastic rod with the cloth.

The plastic rod becomes negatively charged.

(i) Compared with the plastic rod, which row of the table is correct for the charge on the cloth?
Put a cross (☒) in the box next to your answer.

(1)

	sign of charge	size of charge
☐ A	positive	equal
☐ B	negative	equal
☐ C	positive	bigger
☐ D	negative	bigger

(ii) Explain how the plastic rod becomes negatively charged.

(2)

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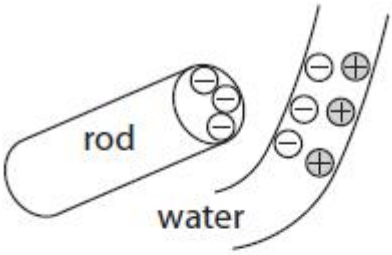
(iii) The student then holds the plastic rod near to a stream of water coming from a tap.

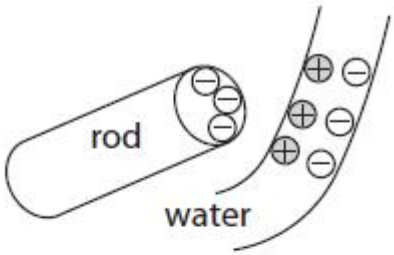
The stream of water bends towards the plastic rod.

Which picture shows the correct arrangement of charges in the stream of water?
Put a cross (☒) in the box next to your answer.

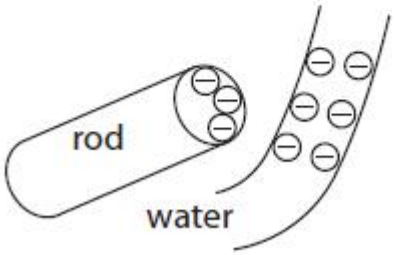
(1)



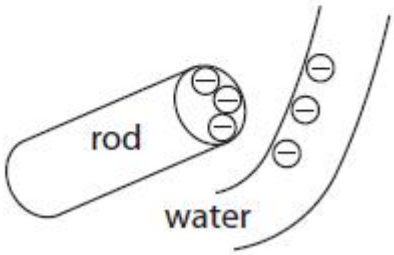
☐ **A**



☐ **B**



☐ **C**



☐ **D**

(iv) The student puts the plastic rod into the stream of water and pulls it out.

Now, when he holds the plastic rod near the stream of water, the stream of water does not bend.
Suggest why the stream of water does not bend.

(1)

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Q4.

(a) A student wins a trophy. It is a metal cup on a black plastic base.

The student cleans the trophy.
She holds one of the metal handles and rubs the rest of the trophy with a dry cloth.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

The plastic base becomes negatively charged because it gains

(1)

A atoms

B becquerel

C einstein

D radium

(ii) Explain why the base gains a negative charge when she rubs the trophy with the cloth.

(2)

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(iii) The metal cup does not become charged when she rubs the trophy.
Suggest why the cup does not become charged.

(2)

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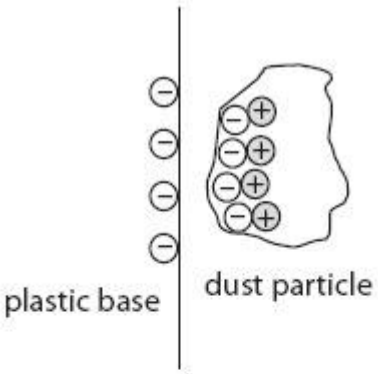
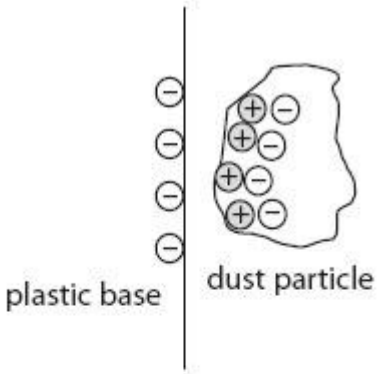
(iv) Some dust particles in the air drift near to the plastic base just after she cleans the trophy.

Which diagram shows the correct distribution of charges on a dust particle near to the charged plastic base?
Put a cross (☒) in the box next to your answer.

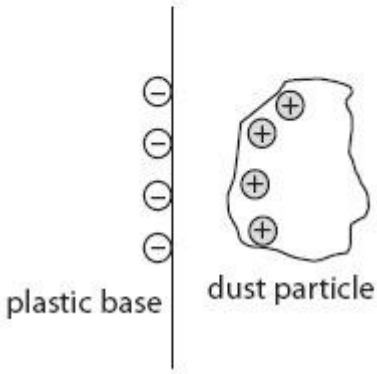
(1)



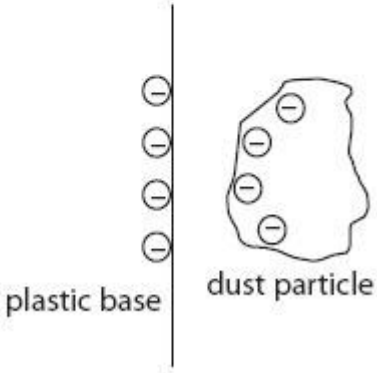
☐ A



☐ B



☐ C



☐ D

(b) Describe **one** situation where separation of electric charge can create a spark.

(2)

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(Total for Question = 8 marks)

Q5.

An atom contains electrons, neutrons and protons.
(a) Use words from the box to complete the sentences.

neutral	negative
much larger than a neutron	much smaller than a neutron
positive	the same size as a neutron

d

(i) The charge on an electron is .....

(1)

(ii) An electron has a mass that is .....

(1)

(b) At a petrol station, a pipe is used to transfer petrol to the storage tanks.

The pipe is earthed.

There is friction between the petrol and the end of the pipe.

(i) Explain why it is dangerous **not** to earth the pipe.

(2)

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(ii) Explain how earthing the pipe makes this process much safer.

(2)

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*(c) The photographs show some electrostatic effects.

positively charged balloon near hair

positively charged rod near some paper

positively charged balloon near a thin stream of water

Explain in terms of electric charges how one of these effects is caused.
You may include diagrams to help with your answers.

(6)

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(Total for Question is 12 marks)

Q6.

State **one** way of giving a plastic rod a static electric charge.

(1)

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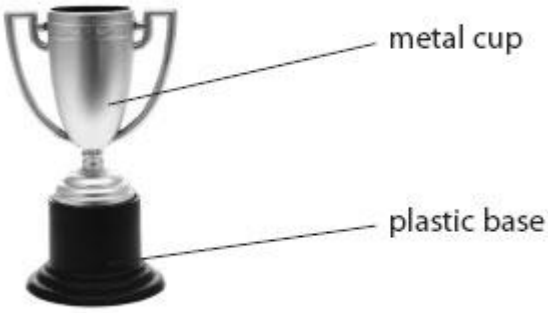
(Total for question = 1 mark)

Q7.

A student wins a trophy. It is a metal cup on a black plastic base.



The student cleans the trophy.
She holds one of the metal handles and rubs the rest of the trophy with a dry cloth.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

The plastic base becomes negatively charged because it gains

(1)

A atoms

B becquerel

C einstein

D radium

(ii) Explain why the base gains a negative charge when she rubs the trophy with the cloth.

(2)

.....

.....

.....

(iii) The metal cup does not become charged when she rubs the trophy.
Suggest why the cup does not become charged.

(2)

.....

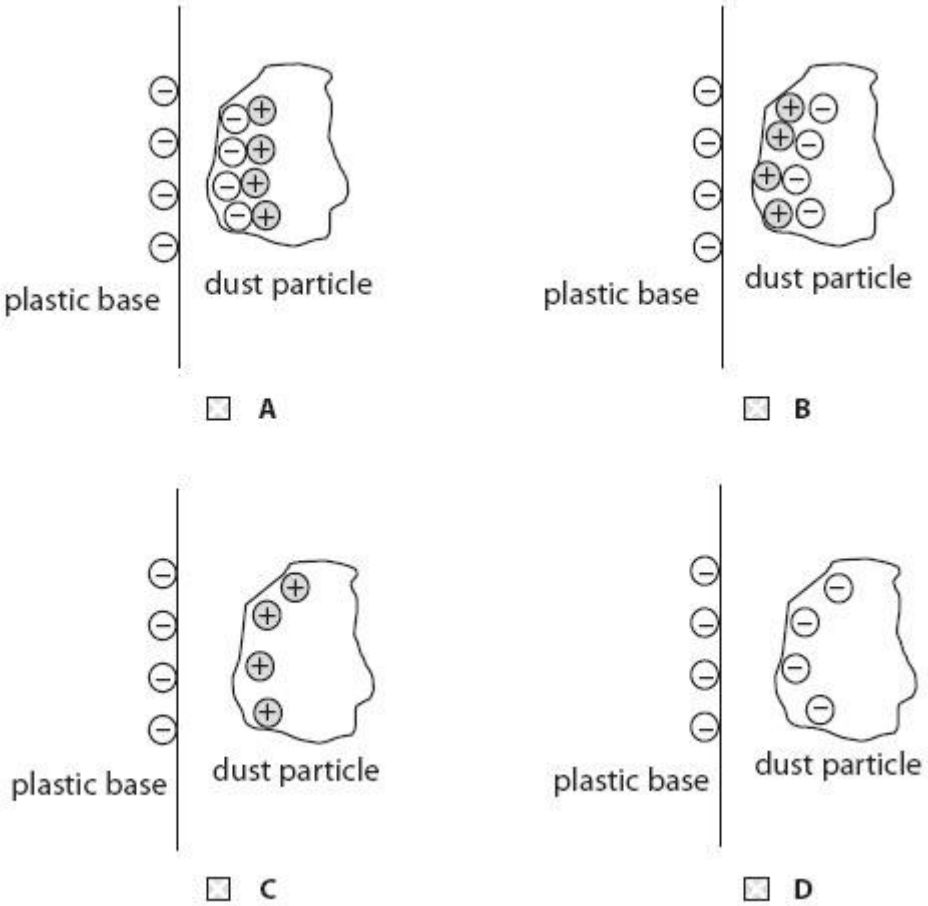
.....

.....

(iv) Some dust particles in the air drift near to the plastic base just after she cleans the trophy.

Which diagram shows the correct distribution of charges on a dust particle near to the charged plastic base?
Put a cross (☒) in the box next to your answer.

(1)



Q8.

The student rubs another balloon with a cloth.
This balloon becomes negatively charged.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

Compared to the charge gained by the balloon, the cloth gains

(1)

☐ **A** a larger negative charge

☐ **B** a larger positive charge

☐ **C** an equal negative charge

☐ **D** an equal positive charge

(ii) Explain why the balloon became negatively charged when it was rubbed with the cloth.

(2)

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(iii) The student then puts this charged balloon against a metal cabinet.

Describe what happens to the charge on the balloon where it touches the metal cabinet.

(2)

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(iv) The student charges another balloon and holds it against a wall. The charged balloon sticks to the wall when he lets go.

Suggest why the balloon is attracted to the wall.

(1)

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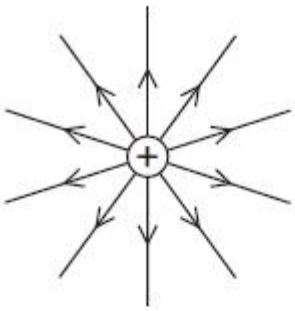
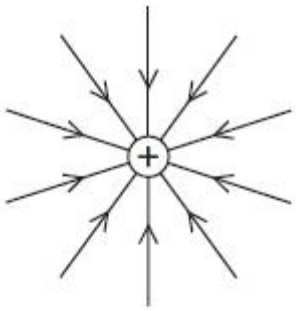
Q9.

Which of these diagrams shows the shape and direction of the electric field around a positive point charge?

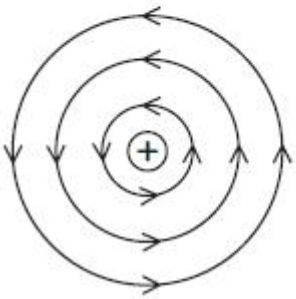
(1)



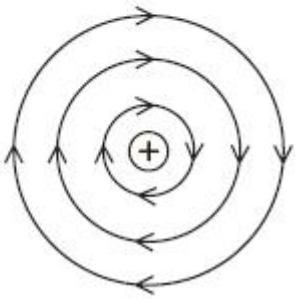
☒ **A**



☒ **B**



☒ **C**



☒ **D**

(Total for question = 1 mark)

Q10.

Balloon **P** hangs from an insulating thread.

A teacher gives the balloon a positive electric charge, as shown in Figure 15.

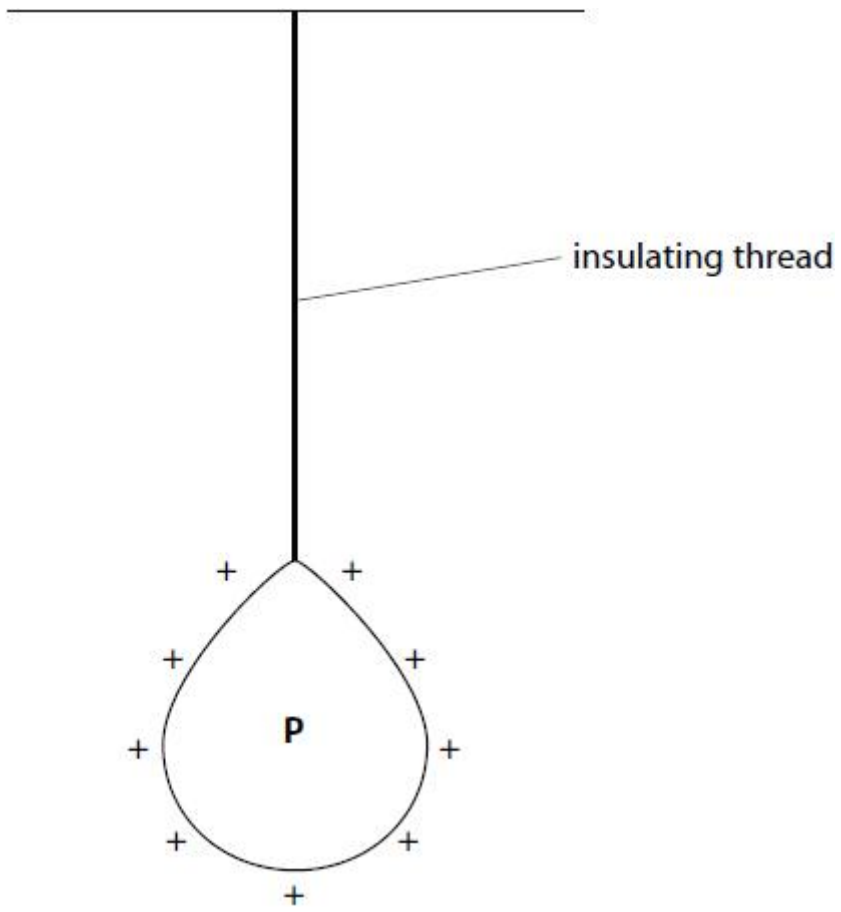


Figure 15

When the balloon is charged like this, it has

(1)

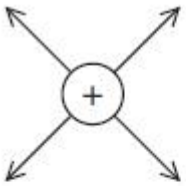
- ☐ **A** gained electrons
- ☐ **B** lost electrons
- ☐ **C** gained protons
- ☐ **D** lost protons

(Total for question = 1 mark)

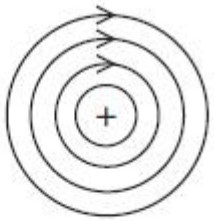
Q11.

Which of the following shows the electric field around a point charge?

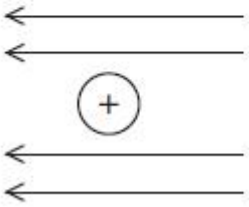
(1)



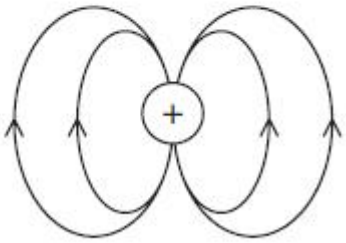
☒ **A**



☒ **B**



☒ **C**



☒ **D**

(Total for question = 1 mark)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	An answer that combines the following points of understanding to provide a logical description: the situation which caused the charge separation (1) where the spark travelled {from/to} (1)	examples when refuelling, spark between end of {fuel/pipe} and vehicle =2 spark {between/from /to} person comb/clothes/metal handle and, when combing hair/removing clothing/opening door = 2 lightning flash, between cloud and cloud/plane/ground, =2 ignore between plug and socket/jump leads	(2)

Question number	Answer	Additional guidance	Mark
(ii)	unit conversion (1) $0.22\ \mu C = 0.22 \times 10^{-6}$ C and $2\ ms = 2 \times 10^{-3}$ s substitution (1) 0.22×10^{-6} = current x 2×10^{-3} s rearrangement (1) current = $0.22 \times 10^{-6} / 2 \times 10^{-3}$ evaluation (1) 1.1×10^{-4} (A)	Substitution and re-arrangement in either order both needed ecf award full marks for correct answer without working power of ten error only loses one mark, if the rest is correct	(4)

Q2.

Question Number	Answer	Acceptable answers	Mark
(a)	repel (1) charge (1) positive (1) electrons (1)		**(4)**

Question Number	Answer	Acceptable answers	Mark
(b)(i)	An explanation linking any **three** from the following: • Droplets have same charge (1) • (droplets) repel (one another) (1) • (This produces) a fine spray/mist (1) • attraction between droplets and plant (1) • This improves coverage OR Spray covers whole [leaf /plant] top and underside of leaf/ gives a fine coating/ even coat (1) • Less spray used/wasted/ falls onto soil (so saves money) (1)	Ignore references to attracting or repelling insects. ignore droplets are positive /negative droplets spread out (produce an) even spray droplets attracted to negative/opposite charge (on plant) or spray will stick to leaves/plant better/more chance of spray landing on/hitting plant or spray (lands) evenly on plant none is wasted/Less will run off the leaves/Same amount of spray will cover a larger area(so saves money)	**(3)**

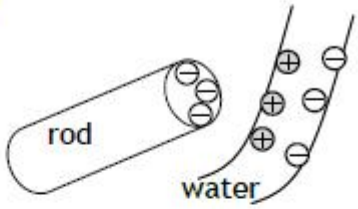
Question Number	Answer	Acceptable answers	Mark
(b)(ii)	10 minutes = 600 seconds (1) substitution 0.008 x 600 (1) evaluation 4.8 (C) (1) Ignore any unit given by the candidate	ECF from their time eg 2 marks for 0.08 if their time is 10 0.8/8/8.0/80 gains 1 mark (bod POT error) Power of ten error max of 2 marks eg 480 gains 2 marks Award 3 marks for correct answer, no working No power of ten error mark if answer less than 0.008 as probably dividing Award 2 marks for 0.08, no working	**(3)**

(Total for Question =10 marks)

Q3.

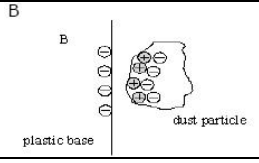
Question Number	Answer	Acceptable answers	Mark
(i)	**A** positive : equal (1)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(ii)	An explanation linking negative charge(s)/electrons (1) (move/ transfer) {to (plastic) rod / to it / from cloth} (1)	Any reference to positive charges, positive electrons or protons moving scores zero marks for question ignore contradictions to Q i.e. cloth is negatively charged attract is insufficient for transfer e.g. {rod /it} gains/gets electrons (from cloth) for 2 marks the cloth loses electrons (to the rod) for 2 marks	**(2)**

Question Number	Answer	Acceptable answers	Mark
(iii)	B 		**(1)**

Question Number	Answer	Acceptable answers	**Mark**
(iv)	a suggestion including: plastic rod has {become neutral/ discharged/no longer charged/not negatively charged (anymore)} OR {charge/electrons} {transferred/ taken} from rod (to/by the water) (1)	Any reference to positive charges, positive electrons or protons moving scores zero marks for question accept the rod loses its charge/ electrons OR rod is 'earthed'/ 'grounded' ignore has same charge as water the water removes/washes away the electrons/charge	**(1)**

Q4.

	Answer	**Acceptable answers**	**Mark**
(a)(i)	B electrons		**(1)**
(a)(ii)	An explanation linking (negative) electrons transfer (1) because of friction/from cloth (to base) (1)	negative charge (reject protons and positive charge for this mp) moves cloth loses electrons/negative charge (to base) = 2	**(2)**
(a)(iii)	A suggestion to include charge (any) could move through cup /metal (1) (cup is) earthed (1)	cup/metal is a conductor ignore metal is not an insulator to earth/ ground / to/ through student's hand	**(2)**
(a)(iv)	B 		**(1)**
(b)	A description to	examples	**(2)**

	include the situation which caused the charge separation (1) where the spark travelled from or to(1)	when refuelling, spark between end of fuel/pipe and vehicle =2 spark between/from /to person comb/clothes/metal handle and, when combing hair/removing clothing/opening door = 2 lightning flash, between cloud and cloud/plane/ground, =2 ignore between plug and socket/jump leads	

Q5.

	Answer	Acceptable answers	Mark
(a)(i)	negative (1)		**(1)**
(a)(ii)	(much) smaller than a neutron (1)		**(1)**
(b)(i)	An explanation linking • (friction/it) produces charges (at the end of the pipe) (1) • charge jumps to fuel tank (1) • (charge/friction) causes a spark (1) • can cause a fire /explosion (1)	static (electricity) builds up	**(2)**
(b)(ii)	An explanation linking • (excess) charge / electrons (1) • Removed/ conducts away (1)	static charge discharged/ neutralised discharge current scores both marks	**(2)**

		Indicative Content	Mark
QWC	***(c)**	An explanation etc. including some of the following points • static electricity • opposites charges attract • charges are different	**(6)**

		• induced charges • charges separate • charges move • electrons move • electrons move towards a positive charge / balloon / rod Allow credit for a correct explanation for an effect which is not given in the question. Allow credit for separation of charge being shown on a diagram.	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited explanation. Explains the effect is caused by charges. e.g. the charge on the balloon pulls the water; the charge on the rod attracts the bits of paper; the balloon is rubbed to give it charge; opposites attract; positive and negative attract; • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple explanation. Explains an effect is caused by opposite charges attracting or like charges repelling. e.g. the charge on the balloon is opposite to the charge on the water so they attract; the positive charges on the balloon attract negative charges on the girl's hair; • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed explanation. Explains the effect is caused by induction, charge separation or moving electrons which leads to attraction between opposite charges. e.g. the electrons have been moved off the balloon so it has a positive charge and attracts the negative charge on the hair; the balloon has a positive charge and induces a negative charge on the stream of water which attracts it; • the answer communicates ideas clearly and coherently uses a range of scientific	

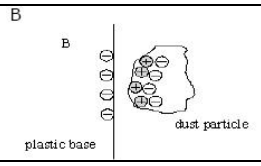
		terminology accurately • spelling, punctuation and grammar are used with few errors

Total marks for question = 12

Q6.

Question number	Answer	Additional guidance	Mark
	rub the rod with (a cloth) (1)	friction	**(1)** **AO1**

Q7.

	Answer	Acceptable answers	Mark
(i)	B electrons		**(1)**
(ii)	An explanation linking (negative) electrons transfer (1) because of friction/from cloth (to base) (1)	negative charge (reject protons and positive charge for this mp) moves cloth loses electrons/negative charge (to base) = 2	**(2)**
(iii)	A suggestion to include charge (any) could move through cup /metal (1) (cup is) earthed (1)	cup/metal is a conductor ignore metal is not an insulator to earth/ ground / to/ through student's hand	**(2)**
(iv)	B  		**(1)**

Q8.

	Answer	Acceptable answers	Mark
(i)	☒ **D** an equal positive charge		**(1)**
(ii)	an explanation linking any two of friction (between	charge/electrons move accept balloon gains	**(2)**

	cloth and balloon) (1) transfer of electrons (1) (electrons/negative charges move) from cloth to balloon (1)	electrons from the cloth for 2 marks	
(iii)	a description including **two** from the following: • balloon becomes discharged (1) • metal /cabinet is a conductor (1) • electrons move through / on to metal / cabinet (1)	earthed / neutral (negative) charge for electrons accept electrons move to earth for 2 marks	**(2)**
(iv)	(surface of) wall (becomes) positively charged /charged by induction (1)	charges on the wall separate charge closest to the surface of the wall is opposite to the charge on the balloon	**(1)**

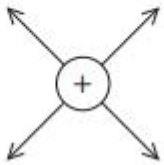
Q9.

Question Number:	Answer	Mark
	A **The only correct answer is A** ***B*** *is not correct because the arrows are in the wrong direction* ***C*** *is not correct because the field is not circular* ***D*** *is not correct because the field is not circular*	**(1)** AO 1 1

Q10.

Question number	Answer	Mark
	B	(1)

Q11.

Question number	Answer	Additional guidance	Mark
	 ☐ **A** B C and D are incorrect because they do not show the electric field around a point charge.		**(1)** **AO1**